

Agraw Mindaye

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TECHNICAL SKILLS

Languages: C, C++, Python, Assembly

Embedded: STM32 (Cortex-M0/M33), ESP32, Arduino, ARM GCC, bare-metal, board bring-up, FatFs

Protocols & Interfaces: I2C, SPI, UART, PWM

Sensors & Hardware: LSM6DSOX (IMU), TMP117, thermistors, logic analyzer, oscilloscope, GDB, SWD

Software & Tools: Git, Linux, Make, STM32CubeIDE

Certifications & Training: Embedded Systems (PyjamaCafe), Android Development (CodePath)

EXPERIENCE

Embedded Firmware Engineer

Dec. 2025 – Present

Stukes Defense

Lyndhurst, NJ

- Leading the development of a secure long-duration data-acquisition platform (STM32U575ZI, ARM Cortex-M33), owning the stack from MCU bring-up through encrypted on-card logging and host-side decode
- Brought up the MCU baseline (80 MHz clock tree, watchdogs, RTC, USB CDC console) and added a boot banner reporting firmware version, device ID, reset cause, and per-peripheral status for field diagnostics
- Authored register-level I2C drivers for an LSM6DSOX 6-axis IMU and a TMP117, gating session start on device-ID probes and exposing a clean sensor abstraction API for upper layers
- Wrote the SD SPI block driver from the protocol up (CMD0-CMD9 init, single/multi-block R/W) and integrated FatFs over FAT32, enumerating SDHC/SDXC media to 31 GB behind a double-buffered RAM manager that keeps card writes off the sample path
- Built a versioned record pipeline with structured IMU and temperature samples packed into 4 KB blocks with 22-byte headers, then Heatshrink-compressed at 1.87× and sealed with AES-256-GCM, producing independently decodable, tamper-evident blocks
- Structured the firmware into layered modules and emitted periodic health-monitoring records (buffer usage, drop counters) and validated the format end-to-end against a host decoder across multi-hour stress testing

AI Trainer

Oct. 2025 – Mar. 2026

Handshake AI

Remote

- Developed and evaluated domain-specific prompts to improve LLM reasoning accuracy across scientific and technical domains
- Analyzed model outputs for correctness, clarity, and logical consistency, identifying failure patterns and documenting findings with structured feedback for internal teams
- Conducted independent research to refine prompt design and evaluation methodologies

Software Development Intern

Sep. 2024 – Dec. 2024

NAGY Ventures

Newark, NJ

- Developed a centralized client portal, streamlining branding and marketing workflows and reducing manual reporting overhead
- Integrated real-time social media APIs to enable dynamic tracking of engagement metrics and performance trends
- Collaborated with cross-functional teams to design a responsive React interface, improving onboarding and client retention
- Improved platform scalability and performance by optimizing component rendering and responsiveness across devices

PROJECTS

Smart Environment Dashboard | ESP32, C++

[\(GitHub\)](#)

- Engineered an ESP32-based embedded system to collect and log temperature and humidity data to an SD card
- Integrated an I2C LCD and physical button interface to enable real-time navigation through stored sensor logs
- Designed non-blocking data logging and update logic to maintain display stability during sensor polling and user input

LED Control Panel | Arduino, C++

[\(GitHub\)](#)

- Developed a hardware control system using button inputs and LED outputs, implementing debouncing and state-based logic for reliable operation
- Structured application logic to separate hardware control from state management, improving modularity and maintainability

EDUCATION

New Jersey Institute of Technology

Newark, NJ

Bachelor of Science in Computer Science

Sep. 2022 – Dec. 2024